

MTH/CSE 4224 — Intro to Machine Learning

Project 1

Proposal Deadline: Feb 26

Project Report Deadline: Mar 16

Presentations: Mar 16-20 (during two classes + additional sessions to be announced)

Project Description

Pursue a classification or regression (or mixed) problem using machine learning methods learned in class or elsewhere.

You will propose a problem and submit a proposal outlining:

- The goals of your study
- Intended data sources
- Initial plans for cleaning/preprocessing
- Models you plan to use

You may work individually or in teams. Expectations will scale with the number of team members.

Proposal Requirements (Max 1 Page)

Your proposal must include:

1. **Goals of your study**
2. **Data sources** and initial plans for cleaning/preprocessing
3. **Initial plans for methods** you will use

- You should test multiple methods

4. **Team information**

- List members and each person's planned focus area

Notes

- Methods not covered in class are allowed, but you must include some class methods for benchmarking.
- If you form a team, you cannot split after proposal approval.

Proposal Grading (15 Points Total)

- 5 points: Clear description of project goals
- 5 points: Appropriate data located or plan to obtain data
- 5 points: Reasonable preprocessing and modeling plan

The purpose of the proposal is to design a feasible project at an appropriate difficulty level.

Project Requirements

All projects should include four main components:

1. Data Preparation

- Create or locate dataset
- Clean and preprocess data

2. Benchmarking

- Initial results using basic, untuned ML methods

3. Training and Tuning

- Train models
- Tune hyperparameters to optimize performance

4. Communication

- Write a detailed report including:
 - Diagrams
 - Decisions made
 - Progress from benchmarks
 - Final findings

- Present your work to the professor

Project Grading (150 Points Total)

- 15 points: Proposal

- **15 points:** Problem description in report
- **20–40 points:** Data preparation
- **50–70 points:** Training and tuning models
- **15 points:** Conclusions and diagrams
- **15 points:** Presentation

Point allocations vary depending on project type.

Data Sources (Examples)

- Google Dataset Search
- UCI Machine Learning Repository
- Kaggle
- Spotify API
- https://www.interviewqs.com/blog/free_online_data_setsLinks to an external site.

You can use many types of data, including:

- Audio
- Images
- Sensor readings
- Text
- Video

Data Guidelines

- Keep dimensionality manageable for computational constraints
- A few thousand dimensions is acceptable
- You may reduce resolution or crop large data

Recommended dataset size:

- At least **1000 data points**
- At least **20 features (dimensions)**